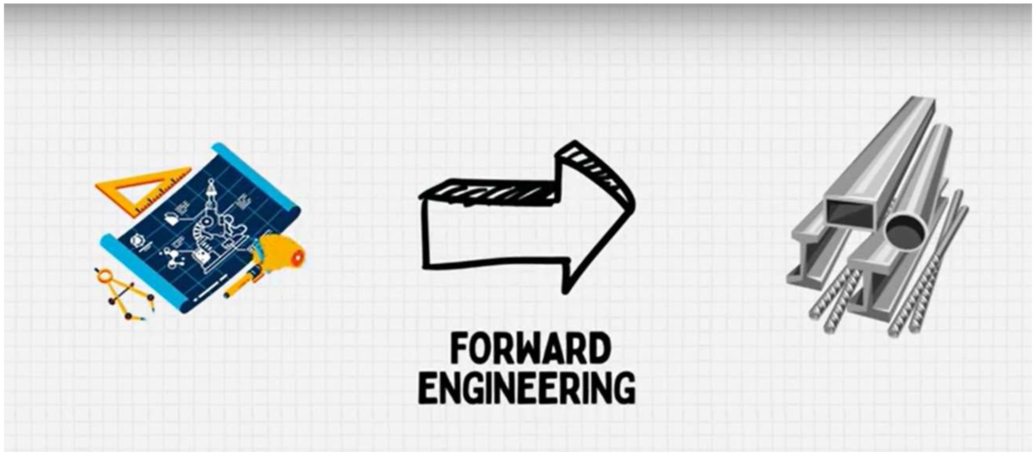
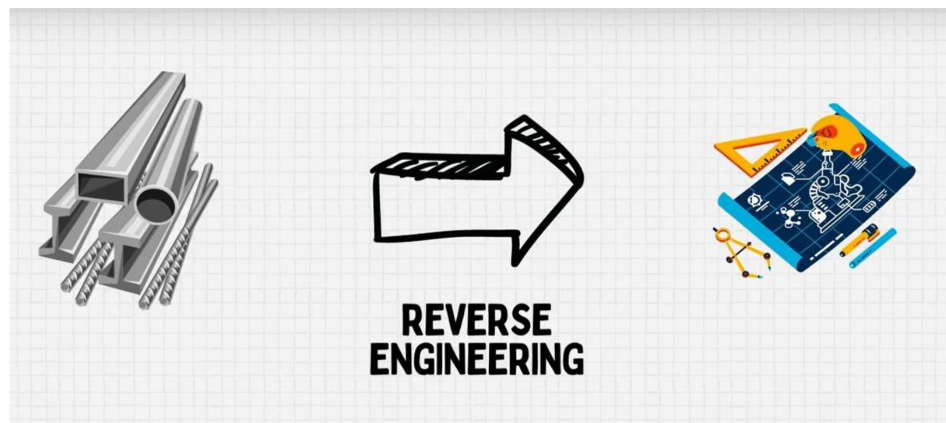
Reverse Engineering


FORWARD
ENGINEERING


REVERSE
ENGINEERING

Introduction to Reverse Engineering

- Reverse Engineering (RE), also known as reverse project, is a description of the product design process. In the general concept of engineering and technical personnel, the product design process is a process conducted from scratch. That is, first designing a concept of the product configuration, performance and generally the technical parameters, and then using CAD technology to build a three-dimensional digital product model.
- It will eventually takes the model into the manufacturing process, to complete the entire design and manufacture of the product cycle. We can call the product design process ‘‘forward design’’.

- Reverse engineering is a process that analyzes the structure of an existing product in order to recreate that product.
- It is especially useful for complex, irregular free-form products. It uses 3-D digital measuring instruments to accurately and quickly measure the outline of coordinates, and to build the surface. After revision and edit, the data is transferred to the general CAD/CAM system.
- The NC tool path generated from the CAM is sent to the CNC machining processing for the required production of the mold, or to a rapid prototyping machine for the production of product models or samples.

Reverse Engineering

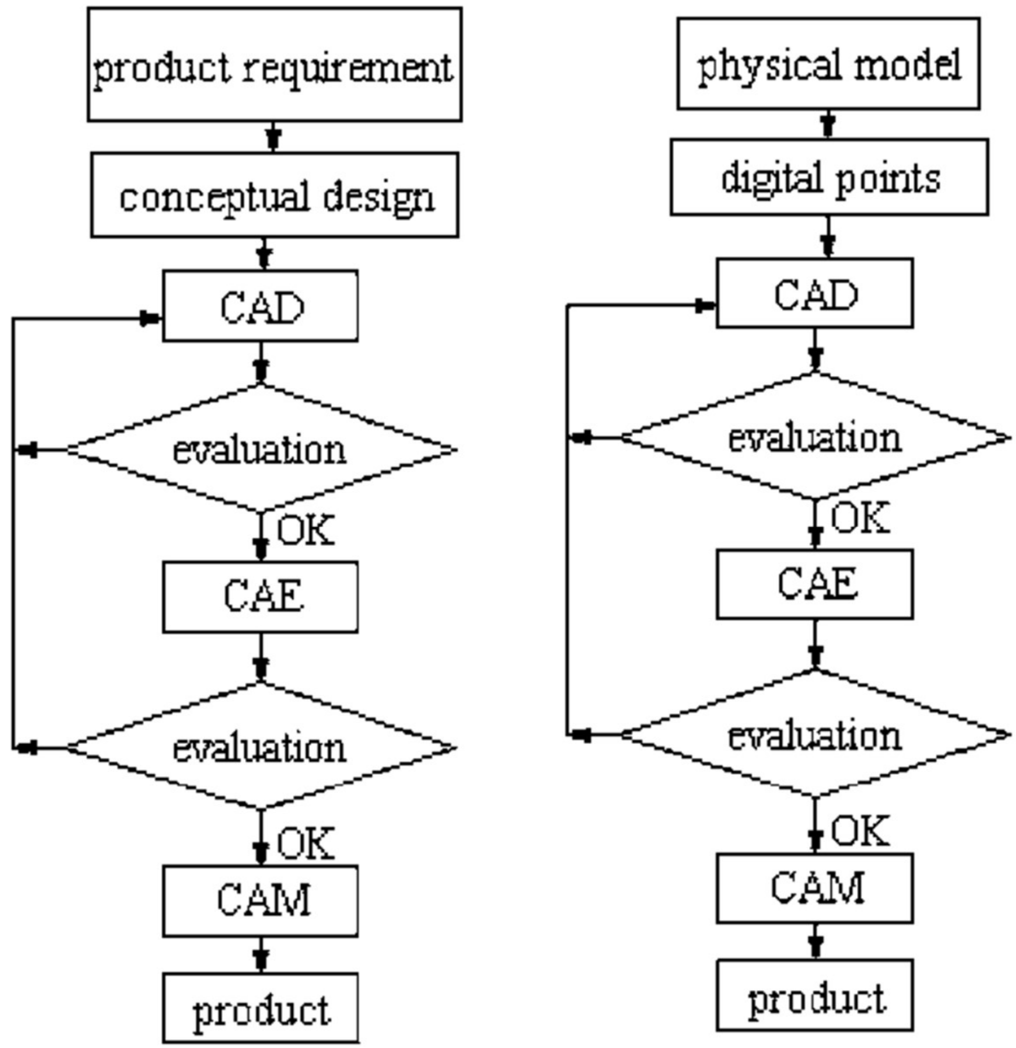
- Reverse Engineering (RE) is a process of obtaining a geometric CAD model from 3-D points acquired by scanning/digitizing existing parts/products.

Reverse Engineering

Importance:

- We cannot start from the very beginning to develop a new product every time.
- We need to optimize the resources available in our hands and reduce the production time keeping in view the customers' requirements.
- For such cases, RE is an efficient approach to significantly reduce the product development cycle.

For example: Impeller Pump Design


product requirement
conceptual design
CAD
evaluation
OK
CAE
evaluation
OK
CAM
product
physical model
digital points
CAD
evaluation
OK
CAE
evaluation
OK
CAM
product

Basic Theory of Reverse Engineering

Reverse engineering consists of three main stages:

- data measurement
- data processing and
- model reconstruction.

Corresponding to these phases of reverse engineering are three basic theories:

data measurement theory, data processing theory and model reconstruction

Is Reverse Engineering legal?

- It is also often lawful to reverse-engineer a product or process as long as it is obtained legitimately.
- If the product is patented, it doesn't necessarily need to be reverse-engineered, as patents require a public disclosure of invention.
- It should be mentioned that, just because a piece of product is patented, that does not mean the entire thing is patented; there may be parts that remain undisclosed.

Is Reverse Engineering legal?

Justifying RE:

- The fundamental use of Reverse Engineering is to get the feel of the product, in terms of dimensional accuracy. This helps in improvisation and to determine the flaws in the product.
- In other words, the reverse engineering process in itself is not concerned with creating a copy or changing the artifact in some way; it is only an analysis in order to deduce design features from products with little or no additional knowledge about the procedures involved in their original production.
- Even when the product reverse engineered is that of a competitor, the goal may not be to copy them, but to perform competitor analysis.

Is Reverse Engineering legal?

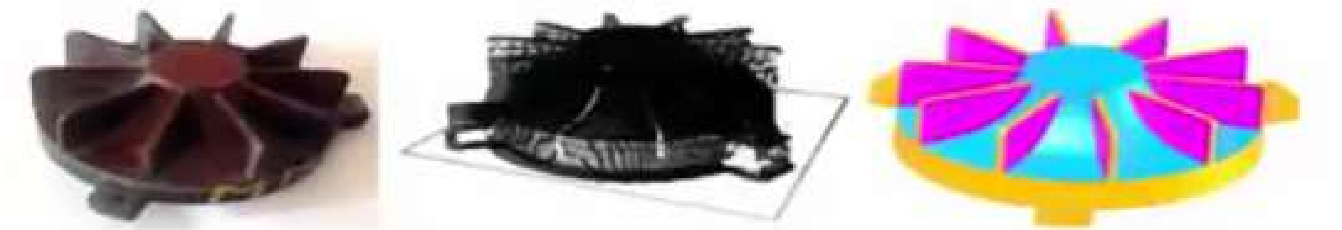
Justifying RE:

1. Interfacing:
2. Military or commercial surveillance:
3. Obsolescence:
4. Product security analysis:
5. Competitive technical intelligence:
6. Saving money:

Reverse Engineering Process

1. Digitization of the object/ Data Capturing (using CMM, scanners etc.)
2. Processing of measured data
3. Creation of CAD model.
4. Prototype.

Reverse Engineering Process



a. b. c.

CAD model generation using laser scanner:
(a) wooden pattern, (b) cloud of points, and (c) 3D CAD model

Scanning

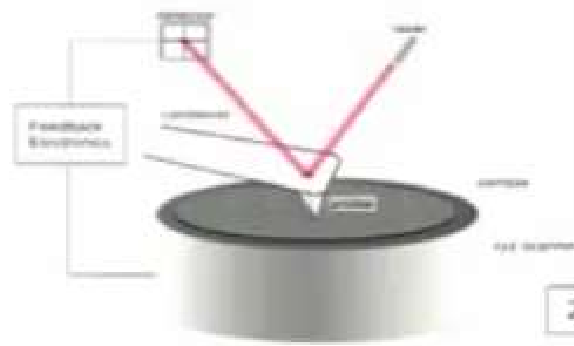
Contact Scanners:

- CMM Based.
- Soft materials can't be scanned accurately.
- Comparatively slow process.

Noncontact Scanners:

- Uses lasers, optics and CCD sensors to capture data points.
- Shiny surfaces and surfaces parallel to light axis can't be scanned accurately.

1



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3D Scanning Process

For an ideal scanning process, the scanning procedure has been divided into five key steps:

1. Acquisition
2. Alignment
3. Mesh Generation
4. Post Processing
5. Simplification

3D Scanning Process

1. Acquisition

- The acquisition is the first fundamental step in which the acquired image is created in the software as a set of points.
- These points define a 3D representation of the part of the object that has been framed and hit by the light pattern generated by the projector.
- For this reason, it is advised to proceed with the acquisition of a wide part of the object first, postponing the acquisition of details and missing parts in a following moment.
- Once a rough 3D reconstruction has been obtained, the scan can be improved by adding more views that correspond to some missing parts.

3D Scanning Process

2. Alignment

- Alignment is the work phase, where, it is possible to bring to the same reference system (align) the Range Images (i.e. images acquired by using the scanner at different lengths from the object) acquired previously.

3D Scanning Process

2. Alignment

Manual Alignment:

- The process is manually helped by the identification of three corresponding points between the two acquisitions taken into account.

Global Alignment:

- Beside the manual alignment, that works with the identification of three corresponding points, another alignment tool called 'global alignment' is also available.
-
- It is advised to run this command after having manually aligned all the range images, in this way the alignment of each acquisition is optimized with respect to the others

3D Scanning Process

3. Mesh Generation

Once a sufficient number of range images has been acquired and aligned in order to create a 3D model as complete as possible, the next step is to generate a triangular Mesh.

- The Mesh generation converts a set of 3D points (Range Image) to a data constituted by a set of triangles (Mesh).
- The Mesh is the first useful data that can be elaborated and exported in the available formats.

3D Scanning Process

4. Post Processing

- Post Processing is every operation that involves the enhancement and finishing of a mesh. Its purpose is to prepare a complete and flawless 3D model ready to be exported.

- These operations should be chosen depending on the result to be achieved and they can affect more or less the 3D model.

3D Scanning Process

4. Post Processing

Make Manifold:

- This involves solving the possible topological issues that can be attributed to the presence of triangle edges shared by more than two faces.

Detect and repair intersection:

- Solving some possible topological issue, attributed to the triangles that intersect other triangles of the mesh surface.

Fill Holes:

- Detection and fixing of missing parts on the mesh surface. It automatically fills the missing data with a surface composed of triangles that propagates the nearby shape and texture information.

3D Scanning Process

5. Simplification

Under this process, are gathered all the steps made on the mesh that (tend to) simplify the data.

Reduce noise on mesh:

- When the generated 3D model presents some surface imperfection, such as roughness or orange peel effect, a filter commonly defined as reduce noise, can be applied to smooth the surface.
- This is an operation works like a digital sandpaper.

3D Scanning Process

5. Simplification

Under this process, are gathered all the steps made on the mesh that (tend to) simplify the data.

Mesh Decimation:

- Reduction of the number of mesh triangles.
- This operation can be done forcing a tolerance that guarantees that the decimated 3D model does not differ more than this value from the original model.
- It is done in order to have a more manageable file that is quicker to elaborate with the post processing tools, and of smaller dimension, thus easier to share on the internet and with less occupancy on the hard disk, without losing the 3D model details.

Reverse Engineering:
Hardware & Software

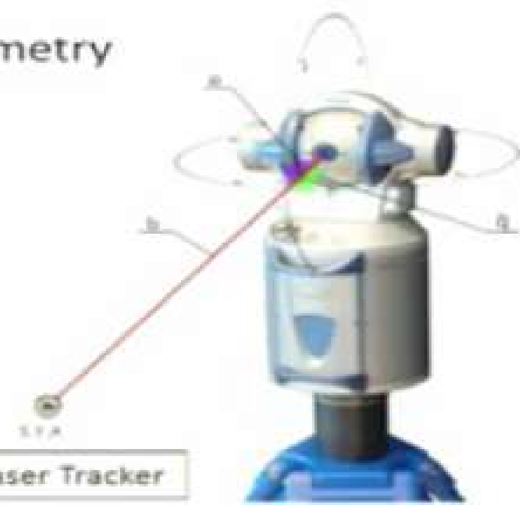
- RE hardware is used for RE data acquisition.
- There are three main technologies for RE data acquisition: contact, non-contact and destructive.
- Outputs of the RE data acquisition are 2-D cross-sectional images and point clouds that define the geometry of an object.
- RE software transforms the RE data produced by RE hardware into 3-D geometric models.
- RE data processing chain can be one of two types of 3-D data: (i) polygons or (ii) curves
- Polygon models are commonly used for rapid prototyping, laser milling, 3-D graphics, simulation, and animations.

Reverse Engineering: Hardware

Classification:

- Laser Trackers
- Total Stations
- Digital Photogrammetry
- Portable CMM's

3. Total Station



2. Laser Tracker

Reverse Engineering: Hardware

Contact Methods:

- Use sensing devices with mechanical arms, CMM and CNC m/c to digitize a surface.
- Data collection techniques:
  1. point to point sensing with touch-trigger probes installed on CMM or articulated mechanical arm to gather co-ordinate points of a surface.

     CMM provides more accurate measurement data compared the articulate arm, but due to lack of no. of DOF, it can't be used to digitize complex surfaces.
  2. analogue sensing with scanning probes installed on CMM or CNC m/c .

     The scanning probe provides a continuous deflection output that can be combined with m/c position to derive the location of the surface.

     The scanning speed in analogue sensing is up to 3 times faster than point- to-point sensing.

Reverse Engineering: Hardware

Point Processing

- Importing the point cloud data.
- Reducing the noise in the data collected.
- Reducing the number of points.
- These tasks are performed using a range of predefined filters.
- Multiple scans are sometimes needed to ensure all the required points are scanned.
- A wide range of commercial software are required.
- The output is a clean, merged, point cloud data set in the most

Reverse Engineering: Hardware

Point-to-point sensing equipment

1.

2.

Zenith 3 CNC CMM

Reverse Engineering: Hardware

Analogue sensing equipment

1. SP25M scanning probes from Renishaw Inc.
2. Roland DGA Corp. MDX-20 scanning and milling machine using Roland Active Piezo Sensor for 3D scanning

Reverse Engineering: Hardware

Advantages:

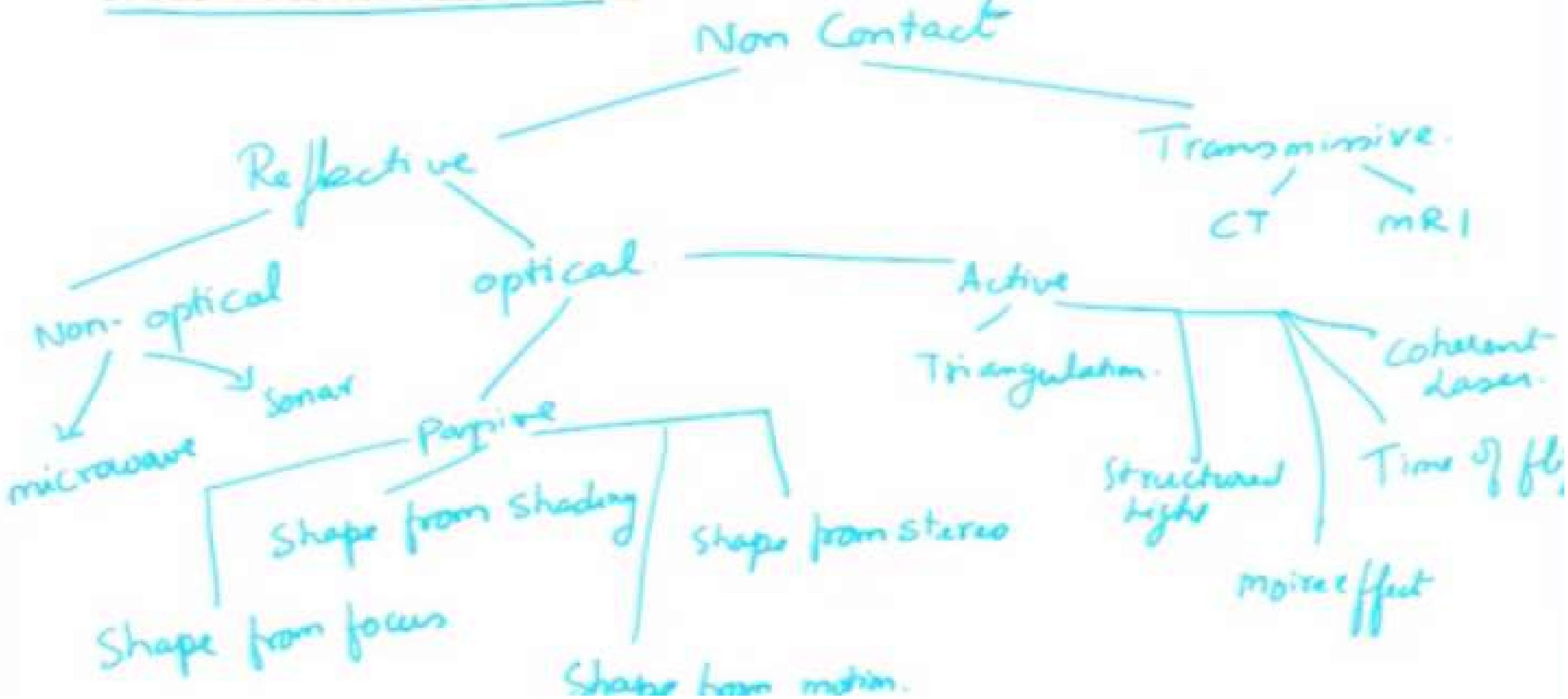
- High accuracy.
- Low costs.
- Ability to measure deep slots and pockets.
- Insensitivity to colour or transparency.

Reverse Engineering

- 2D cross sectional images and point clouds are captured projecting energy sources (light, sound, or magnetic field) on an object.
- Then, either the transmitted or the reflected energy is observed.
- The geometrical data are calculated by using triangulation, time-of-flight, wave interference information, and image processing algorithms.
- The classifications of non-contact RE hardware are based on the sensor technologies or data acquisition techniques employed.

Reverse Engineering:

Non-contact method



Reverse Engineering:
Non-contact method

Advantages of non-contact mode:

- No physical contact.
- Fast digitizing of substantial volumes.
- Good accuracy and resolution for common applications.
- Ability to detect colours.
- Ability to scan highly detailed objects, where mechanical touch probes may be too large to accomplish the task.

Optical techniques

Triangulation

- Most laser scanners use straightforward geometric triangulation to determine the surface coordinates of an object.
- Triangulation is a method that employs locations and angles between light sources and photosensitive devices (CCD) to calculate coordinates.

Optical techniques

Triangulation

- There are two variants of triangulation schemes using CCD cameras: single and double CCD camera.

Single camera system:

- A device transmits a light spot (or line) on the object at a defined angle. A CCD camera detects the position of the reflected point (or line) on the surface.

Double camera system:

- Two CCD cameras are used. The light projector is not involved in any measuring functions and may consist of a moving light spot or line, moving stripe patterns, or a static arbitrary pattern.

Optical techniques

- In the illustration above, a high energy light source is focused and projected at a pre-specified angle (θ) onto the surface of an object.
- A photosensitive device senses the reflection from the illuminated point on the surface.
- Fixed baseline length (L) between the light source and the camera is known from calibration.

Optical techniques

Using geometric triangulation from:

- the known angle (θ),
- the focal length of the camera (F),
- the image coordinate of the illuminated point (P), and
- fixed baseline length (L),
- the position of the illuminated point (Pi) with respect to the camera coordinate system can be calculated as follows:

Optical techniques: Structured Light

- A light pattern is projected at a known angle onto the surface of interest and an image of the resulting pattern, reflected by the surface, is captured.
- The image is then analyzed to calculate the coordinates of the data point on the surface.
- A light pattern can be:
  - a single point;
  - a sheet of light (line) or
  - a strip, grid, or more complex coded light.

Optical techniques: Structured Light

- The most commonly used pattern is a sheet of light that is generated by fanning out a light beam.
- When a sheet of light intersects an object, a line of light is formed along the contour of the object.
- This line is detected and the X, Y, Z coordinates of hundreds of points along the line are simultaneously calculated by triangulation.

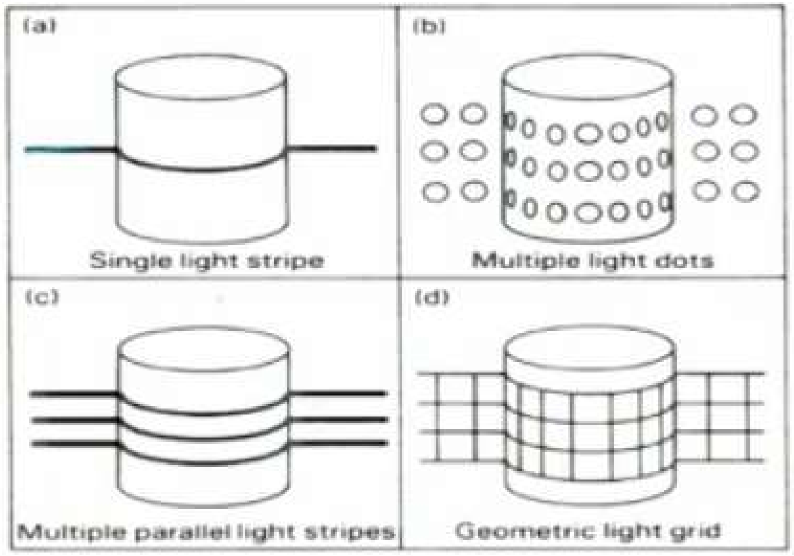
Optical techniques: Structured Light

Features of Structured Light

- The data acquisition is very fast (up to millions of points per second).
- Colour texture information is available.
- Structured-light systems do not use a laser.

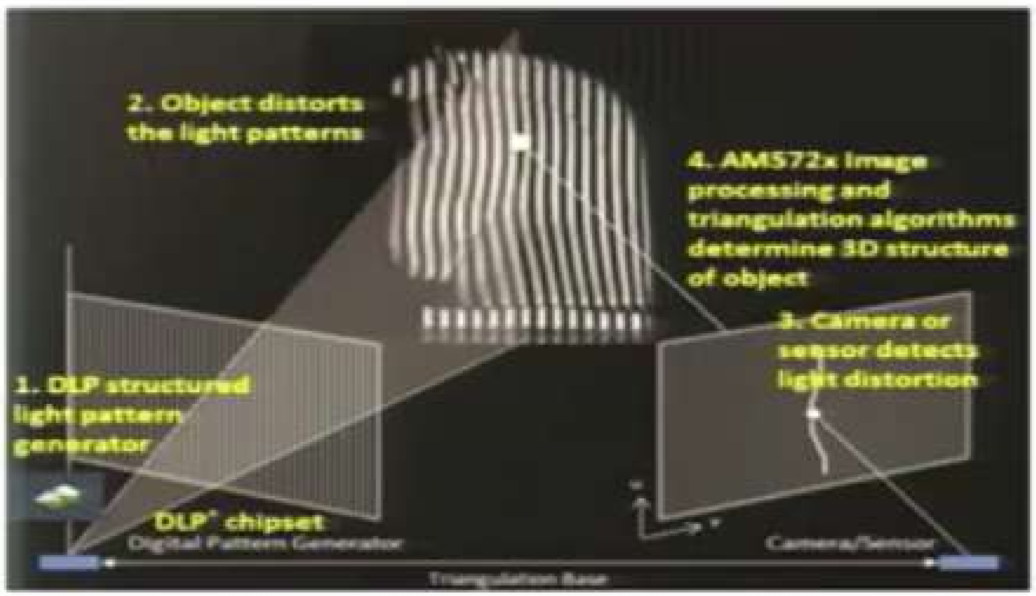
Optical techniques: Structured Light

Different light patterns used in structured-light techniques



Optical techniques: Structured Light

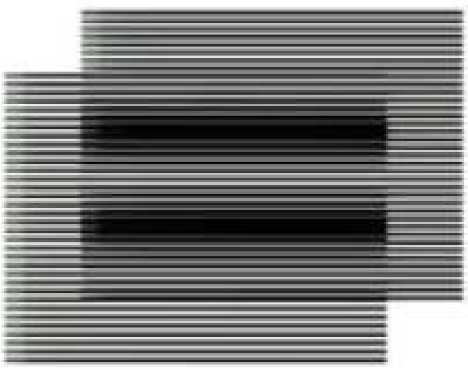
2-D Image acquisition using structured light, parallel light strips pattern.



Optical techniques:
Interferometry (Moiré Effects)

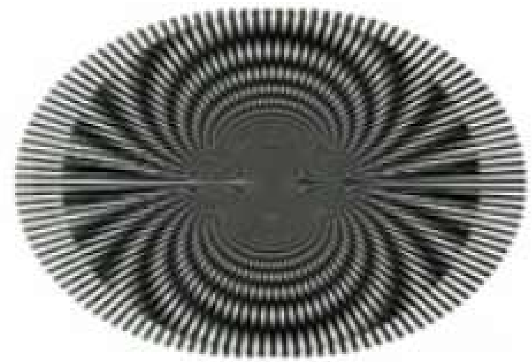
- Used for dimensional inspection and flatness and deformation measurements.
- Structured-light patterns are projected onto a surface to produce shadow Moiré effects.
- The light contours produced by Moiré effects are captured in an image and analyzed to determine distances between the lines.
- This distance is proportional to the height of the surface at the point of interest, and so the surface coordinates can be calculated.
- The Moiré technique gives accurate results for 3-D reconstruction and measurement of small objects and surfaces.

Optical techniques:
Interferometry (Moiré Effects)



1.

2. A moiré pattern, formed by two sets of parallel lines, one set inclined at an angle of 5° to the other.



3. Concentric Circles Pattern

Optical techniques:
Time of Flight

- The principle behind TOF implementations is to measure the amount of time (t) that a light pulse (i.e., laser electromagnetic radiation) takes to travel to the object and return.
- The distance (D) of the object from the laser would then be equal to approximately one half of the distance the laser pulse travelled: $D = C \times t/2$, where C is speed of light.

Optical techniques:

Time of Flight - features

- The high velocity of light allows TOF scanners to make hundreds, or even thousands of measurements per second.
- TOF techniques can digitize large, distant objects such as buildings and bridges.
- The accuracy depends on the pulse width of the laser, the speed of the detector, and the timing resolution.
- The shorter the pulse and the faster the detector, the higher the accuracy of the measurement.
- The main disadvantage is that TOF scanners are large and do not capture an object's texture, only its geometry.
- They are not practical for fast digitization of small and medium-sized objects.

Optical techniques:

Time of Flight

- Passive methods reconstruct a 3-D model of an object by analysing the images to determine coordinate data.
- There is no projection of light sources onto the object for data acquisition.
- The typical passive methods are shape from shading and shape from stereo.
- **Shapes From Shading (SFS)** methods are used to reconstruct a 3-D representation of an object from a single image (2-D input) based on shading information.

Passive Methods

Disadvantages of Shapes From Shading (SFS)

- The shadow areas of an object cannot be recovered reliably because they do not provide enough intensity information.

- The method cannot applied to general objects because it assumes that the entire surface of an object has the same reflectance.

- The method is very sensitive to noise because the computation of surface gradients is involved.

Passive Methods: Shape from Stereo

- It refers to the extension of SFS to a class of methods that use two or more images from different viewpoints for shading based 3-D shape recovery.
- Normally, two cameras are coordinated to generate 3-D information about an object by automatically finding corresponding features in each of the two images.
- Then triangulation is used to measure the distance to objects containing these features by intersecting the lines of sight from each camera to the object.
- Compared to SFS methods, there is improved accuracy. However, finding correspondence between images is extremely difficult and can produce erroneous results from mismatches.

Non-optical techniques

- It includes acoustic and microwave radar.
- The principle of 3-D reconstruction is measuring distances from the sensing device to objects, by measuring the time delay between the transmitted and returned signals.
- Sonar techniques are normally used in 3-D underwater mapping, which provide mariners with a major advancement in obstacle avoidance and navigation.
- Acoustic interference or noise is often a problem, as is determining the focused point location.
- Radar is typically intended for use with long-range remote sensing, especially in airline applications.
- In 3-D reconstruction applications, radar is used to measure distances and map geographic areas and to navigate and to fix positions at sea.

Transmissive techniques

- Computerized tomography (CT) is a powerful transmissive approach for 3-D reconstruction.
- It has also been called computerized axial tomography (CAT), computerized trans-axial tomography (CTAT), digital axial tomography (DAT).
- CT is a non-destructive method that allows 3-D visualization of the internals of an object.
- It provides a large series of 2-D X-ray cross-sectional images taken around a single rotational axis.
- Magnetic resonance imaging (MRI) is a state-of-the-art imaging technology that uses magnetic fields and radio waves to create high-quality, cross-sectional images of the body without using radiation.
- Compared to CT, MRI gives superior quality images of soft tissues in many parts of the body.

Destructive methods

- The RE destructive method is useful for small and complex objects, in which both internal and external features are required to be scanned.
- A CNC milling machine exposes 2-D cross-sectional (slice) images, which are then gathered by a CCD camera.
- The scanning software automatically converts the digital bitmap image to edge detected points, as the part is scanned.
- To remodel the part, 2-D slice images of the part are gathered by destroying the part layer-by-layer.